User Manual

FX Vanguard

Contents

1 Introduction

In the globalized market environment, exchange rate fluctuations play a critical role in the decision-making processes of enterprises, financial institutions, and investors. Traditional exchange rate forecasting methods often struggle to fully capture the complexity and volatility of the market. The intelligent forex risk forecasting model presented in this manual leverages advanced machine learning technologies, combining NLP-based text analysis, multi-dimensional data modeling, and real-time market dynamics to provide more accurate forex risk warnings and trend predictions.

This system is capable of detecting early risk signals in the market and conducting in-depth analysis based on multi-source data, helping users optimize decisions and reduce the uncertainty caused by currency fluctuations. Whether multinational corporations, financial institutions, or forex traders, all can enhance their market sensitivity and improve risk management efficiency through this model.

This user manual will provide a detailed explanation of how to use the model and best practices, serving as a comprehensive guide for your operations.

2 README

2.1 Folder Description

The executable files and operational documents include: a project report, a business plan document, a technical document, a work originality declaration document, a back-end model creation document, a front-end packaging file, a back-end packaging file, iOS APPdemo video, html web page demo video, and this user manual.

2.2 Data Description

The datasets and related files involved in this study are as follows, aimed at providing comprehensive and structured data support:

- `FX.zip`

  This archive contains historical exchange rate data. After extraction, users can access historical exchange rates for multiple currency pairs (including onshore and offshore RMB, USD, JPY, EUR, etc.), as well as related macroeconomic indicators such as the Consumer Price Index (CPI), Economic Policy Uncertainty (EPU), Global Political Risk (GPR), and Gross Domestic Product (GDP).

- `all_currency.xlsx`

  This file contains a professionally organized daily exchange rate dataset for multiple currency pairs, including USD/JPY, USD/CNH, JPY/EUR, JPY/CNY, EUR/USD, EUR/JPY, and EUR/CNH, with open, high, low, and close prices.

- `DATA.xlsx`

  This file includes daily exchange rate data for EUR, JPY, and CNY against USD, detailing the open, high, low, and close prices of each pair.

- `All_Data_SQL.zip`

  This archive contains the full dataset organized into an SQL database for model training and validation. After extraction, users can open and view the data in a SQL environment for further analysis and model development.

2.3 Search Instructions

This platform provides a built-in DeepSeek AI assistant designed to answer forex-related questions in real time and assist with financial decision-making. You can access the assistant in two ways:

- **From the Home Page (iOS)**: On the bottom right corner of the home page, a blue floating icon is available. Clicking it will open a search box where you can input your question and receive a real-time response from the assistant.
- **From the Forex Data Search Page**: Once on this page, you do not need to set an API key. The backend has already connected to the necessary service. Simply enter your question in the search box to begin the Q&A session.

Both methods provide quick and accurate forex insights to enhance your user experience.

2.4 Usage Instructions

2.4.1 HTML Users

If you wish to generate trading strategies via the platform, you may input your desired holding period. The backend model will then generate a corresponding trading strategy and provide trading signals, where:

- 1 indicates a buy signal,
- -1 indicates a sell signal,
- 0 indicates no action.

2.4.2 iOS App Users

Due to current technical limitations, the iOS platform does not yet support custom trading strategies. Instead, users can choose from the following preset strategies:

- Conservative Strategy: Low risk, short-term trades, strict stop-loss rules.
- Moderate Risk Strategy: Balanced approach with moderate risk and return.
- Aggressive Strategy: More trading opportunities and longer holding periods.
- Drawdown-Optimized Strategy: Focused on controlling maximum drawdown.
- Long-Term Trend Strategy: Long-term holdings aiming for trend-based returns.

Users are encouraged to choose a strategy that aligns with their risk preferences and trading objectives.

3 MySQL Database Operations

3.1 Downloading MySQL

- Visit the official MySQL website: `https://dev.mysql.com/downloads/mysql/`
- Select the version suitable for your operating system:
  1. Choose **MySQL Community Server** (free and open-source).
  2. Click **"No thanks, just start my download."** to skip the registration step.
- Download the installer (ensure it matches your system architecture).

3.2 Running MySQL

- For Windows, navigate to the MySQL `bin` directory:

  ```
  cd C:\Program Files\MySQL\MySQL Server 8.0\bin
  ```

- Install MySQL service:

  ```
  mysqld -install
  ```

- Start MySQL service:

  ```
  net start mysql
  ```

3.3 Creating a Database

- In MySQL, use the following SQL command to create a new database:

  ```
  CREATE DATABASE my_database;
  ```

- Then create a simple table (example):

  ```
  CREATE TABLE users (
      id INT AUTO_INCREMENT PRIMARY KEY,
      username VARCHAR(50) NOT NULL,
      password VARCHAR(50) NOT NULL,
      created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP
  );
  ```

- Insert sample data:

  ```
  INSERT INTO users (username, password) VALUES ('alice', 'password123');
  ```

3.4 Connecting MySQL in VS Code

- Install MySQL Extension:

  Open vsCode, go to the Extensions Marketplace, search and install the extension **"MySQL for Visual Studio Code"**.

- Configure Connection:

1. Open the MySQL extension from the sidebar.
2. Click "Add Connection" and enter the following details:

```
Host:  localhost
Port:  3306
User:  root
Password:  your MySQL password
Database:  the database name (e.g., my_database)
```

- Connect to the Database:

  Click "Connect" to establish a connection. Once connected, you can execute SQL queries directly within VS Code.

4 Running the frontend files

4.1 iOS APP system

This app is not yet online. If you want to experience it in advance, you can open the project using Xcode. After opening it, you can view the app content in the preview window by clicking on any SwiftUI file.

4.2 HTML page

4.2.1 Unzip the frontend files

- After extracting the folder of the `FX Vanguard`, find the front-end packaging file in it.
- Extract the front-end packaging file to find the file of the `html.zip` in it, and use a common compression software (such as WinRAR, 7-Zip) or command line tools (such as tar, unzip) to extract it.
- After decompression is complete, go to the generated directory and check that the HTML files and other related resources are complete and intact to ensure the correctness and usability of the frontend files.

4.2.2 Browsing the webpage

- Locate the file `index.html` in the folder after decompression.
- Double-click to open the file, and you can browse the web content in the default browser.

5 Modifying frontend files

5.1 HTML pages

5.1.1 Installation project dependency instructions

To modify the frontend project, open the project file using a code editing tool (such as Visual Studio Code), and you can edit the HTML file.

- CSS file dependencies

  CSS files can be imported directly into the HTML file using the <link> tag:

  ```
  <link rel=stylesheet href=styles.css>
  ```

- JavaScript file dependencies
  - JavaScript files can be imported directly into the HTML file using the <script> tag:

    ```
    <script src=script.js></script>
    ```

  - If you need to use a modular tool, you can install the dependency via npm or yarn and import it in the JavaScript file:

    ```
    npm install chart.js
    ```

- External API dependencies

  1. You can get the API key from the API provider:

     `const apiKey = your_api_key_here;`

     item Initiate HTTP requests using the Fetch API or other libraries such as Axios:

     `fetch(‘https://api.example.com/data?api_key=`$apiKey$`‘) .then(response => response.json()) .then(data => console.log(data)) .catch(error => console.error(Error:, error));`

5.2 iOS APP system

In order to modify the code in the iOS APP system, users need to use Apple's official development tool, the **Xcode** application (this application only supports Apple users). The following are the detailed steps:

1. Installing **Xcode**:

   Users need to go to the Mac App Store or the Apple Developer website at https://developer.apple.c to download and install the latest version of Xcode.

2. Launch the Xcode application and open the project folder fxrisk.

3. After opening, users can modify the code.

6 Modifying Frontend Files

6.1 HTML Page

6.1.1 Installing Project Dependencies

To modify the frontend project, use a code editor such as Visual Studio Code to open the project files. You may then directly edit the HTML files.

- **CSS File Dependencies**

  You can include a CSS file directly in the HTML using the `<link>` tag:

  ```
  <link rel="stylesheet" href="styles.css">
  ```

- **JavaScript File Dependencies**

  - You can include JavaScript files directly using the `<script>` tag:
    ```
    <script src="script.js"></script>
    ```
  - If you are using module bundlers or build tools, you may install dependencies using `npm` or `yarn`, and import them in your JavaScript code:
    ```
    npm install chart.js
    ```

- **External API Dependencies**

  1. Obtain an API key from the service provider:
     ```
     const apiKey = 'your_api_key_here';
     ```
  2. Use Fetch API or libraries such as Axios to make HTTP requests:
     ```
     fetch(`https://api.example.com/data?api_key=${apiKey}`)
       .then(response => response.json())
       .then(data => console.log(data))
       .catch(error => console.error('Error:', error));
     ```

6.2 iOS App System

To modify code in the iOS app system, users need to use Apple's official development tool, **Xcode** (macOS only). Follow the steps below:

1. Install **Xcode**:

   Download and install the latest version of **Xcode** from the Mac App Store or from the official Apple Developer website: `https://developer.apple.com/xcode/`.

2. Launch the **Xcode** application and open the project folder named `fxrisk`.

3. Once opened, users may proceed to edit the source code directly.

7 Running Backend Files

7.1 Running `fxrisk.py`

1. Locate the two compressed folders `muti.zip` and `FX.zip` from the attachments. Extract both files and place them on the desktop.
2. Open the `muti` folder on the desktop and find the `fxrisk.py` script.
3. Before running the script, install the following Python libraries:

```
pip install pandas
pip install numpy
pip install pandas_ta
pip install scikit-learn
pip install xgboost
pip install matplotlib
pip install seaborn
pip install openpyxl
```

4. Users also need to perform the following operations:
   - Replace the original path in the script `/Users/kittentang/Desktop/muti/fxrisk.py` with `./Desktop/muti/fxrisk.py`.
   - Create a folder named `results` on the desktop.
5. After completing these steps, run the `fxrisk.py` file.

7.2 Running `strategy.py`

1. Place both the `strategy.py` and `DATA.xlsx` files on the desktop.
2. Before executing `strategy.py`, install the required Python libraries:

```
pip install pandas
pip install numpy
pip install matplotlib
pip install seaborn
pip install tensorflow
pip install scikit-learn
pip install pandas_ta
pip install openpyxl
```

3. Once installation is complete, run the `strategy.py` file.

7.3 File Path Requirements

To ensure proper program execution, please strictly adhere to the following file path specifications:

1. **Fixed File Location:** Users must not change the location of files. The `fxrisk.py` file **must be located in the same directory as** `process_plot_data.py`, otherwise the program may fail to run. Always maintain the original folder structure to avoid path errors that prevent modules or data files from being recognized.

2. **Consistent Directory Structure:** It is recommended to manage all related files within a designated working directory to maintain a clear and functional directory layout.

7.4 Dependency Installation

```
# Recommended: Use Tsinghua Mirror for faster access
pip install -i https://pypi.tuna.tsinghua.edu.cn/simple \
numpy==1.21.6 tensorflow==2.8.0 scikit-learn==1.0.2 \
pandas==1.3.5 shap==0.41.0
```

7.5 Environment Configuration

```
# Set database environment variables
export DB_HOST='localhost'
export DB_PORT='3306'
export DB_USER='fx_user'
export DB_PASSWORD='your_password'
```

8 UI Operation Guide

This section introduces the platform's functional modules and operation instructions for both HTML and iOS App users.

8.1 HTML User Interface

8.1.1 Homepage

This module is designed to help users gain a deeper understanding of market dynamics and uncover the underlying logic behind exchange rate fluctuations, enabling more informed investment decisions.

- **Current Exchange Rate**
  - Real-time display of the current exchange rate for the selected currency pair.
  - Users can manually refresh the exchange rate data by clicking the refresh button.
- **Risk Signal Intensity**
  - Users can click to view the current risk signal intensity.
- **Market Trends**
  - Exchange rate trends are displayed in line chart format. Users can view more detailed time ranges or trend specifics.
- **Market Overview**
  - Users can view market metrics such as volatility, trading volume, and trend direction.
- **Risk Warning**
  - Users can view detailed risk analysis scores for each currency pair.

8.1.2 Market Analysis Page

This module helps users understand market dynamics and uncover the underlying logic behind exchange rate fluctuations.

- **Impact Factor Analysis**
  - Users can view the contribution of various factors to exchange rate fluctuations. Through quantitative analysis, the impact of each factor on market volatility is visually presented.
- **Market Sentiment**
  - Users can view the proportion of bullish, neutral, and bearish sentiment, gaining real-time insight into market participants' emotional trends.

- **Key Events**
  - Users can view recent key events affecting the market, along with detailed descriptions and their level of impact.
- **Technical Indicators**
  - Users can view current market technical indicators, including RSI, MACD, and Bollinger Bands.

8.1.3 Risk Signal Page

This module processes historical or real-time financial data to automatically generate exchange rate risk signals.

- **Parameter Settings**
  - **Currency Pair:** A dropdown menu is provided for users to select different currency pairs (e.g., USD/CNY, EUR/USD) to specify the analysis target.
  - **Maximum Drawdown:** A slider allows users to set the desired maximum drawdown percentage.
  - **Trading Period:** A numeric input box enables users to define the trading period (e.g., 14 days).
- **Risk Signal Strength**
  - Users can click to view the current risk signal strength.
- **Historical Signal Trend**
  - Users can view the historical trend of risk signals over a specified period (e.g., the past 30 or 90 days).
- **Forecast Analysis**
  - Users can view 24-hour risk forecasts, including the risk score, confidence level, and predicted risk grade.
- **Factor-Based Risk Alerts**
  - Users can click to view the fluctuation trends and impact levels of various market risk factors.

8.1.4 Backtesting Results Page

This module displays the performance of risk signals under various market conditions based on backtesting results.

- **Backtesting Period**
  - Users can select different backtesting periods (e.g., 1 year).
- **Yield Curve**

  - Users can view the yield curve to understand the profit/loss changes over the backtesting period.

- **Key Metrics**
  - Users can view key performance indicators such as annualized return, maximum drawdown, Sharpe ratio, and win rate.

- **Trade Statistics**
  - Users can view statistics such as total number of trades, profit-loss ratio, and average holding period.

- **Risk Analysis**
  - Users can view the analysis results for different categories of risk.

8.1.5 Multi-Currency Pair Analysis Page

This module analyzes multiple currency pairs based on their correlation, helping to generate risk signals and related insights.

- **Currency Pair Correlation Matrix**
  - Users can view the coefficients in the correlation matrix of different currency pairs.

- **Portfolio Analysis**
  - Users can view risk indicators such as portfolio volatility and 1-day 95% Value at Risk (VaR).

- **Currency Pair Weight Distribution**
  - Users can define the weights of different currency pairs in the portfolio.

- **Comprehensive Risk Signal**
  - Users can view the risk signals for various currency pairs.

8.1.6 FX Data Search Page

- **Search Bar**
  - Search Box: Users can input queries such as USD/CNY exchange rate trend analysis or EUR exchange rate forecast.
  - Search Button: Users click to initiate the search operation.

- **Search Result Display Area**
  - Result List: Search results are displayed in list format. Each result item contains the following key information: title (indicating the core content of the result) and reason (explaining the match basis).

8.2 iOS App User Interface

8.2.1 Home Page

- **Currency Pair Selection and Exchange Rate Data**
  - Users can swipe left to select different currency pairs. The interface will display the current exchange rate of the selected pair in real time.
  - Users can manually refresh the exchange rate data by clicking the refresh button.
- **Market Trend Forecast**
  - Exchange rate trends are shown in the form of line charts. Users can explore more detailed time intervals or trend details.
- **Market Probability Analysis**
  - Users can switch tabs or swipe views to explore probability analysis of currency pairs.
- **Risk Alert Popup**
  - Users can view detailed risk assessment scores for selected currency pairs.
- **Market Alerts**
  - The popup contains a "Close" button, which users can click to dismiss the alert. The popup also supports quick navigation to relevant analysis pages for more information.
- **AI Assistant ZhiXiaoHui**
  - The assistant is located in the bottom-right corner. When clicked, it opens a chat window that provides real-time consultation services, answering user questions related to market analysis, trading strategies, and more.

8.2.2 Market Analysis Page

This module helps users understand market dynamics and uncover the underlying logic of exchange rate fluctuations.

- **Top Navigation Bar**
  - **Back Button:** Located at the top left corner. Users can click it to return to the previous page.
  - **Import Data Button:** Located at the top right corner. Users can click to import external data files (e.g., CSV, Excel) for further analysis and processing.
- **Factor Impact Analysis**

- Users can view the contribution ratio of various factors to exchange rate fluctuations. The platform provides a visual and quantitative analysis of each factors impact on market volatility.

- **Market Sentiment**
  - Users can view the proportions of bullish, neutral, and bearish market sentiment to understand participants current attitudes.

- **Fear and Greed Index**
  - Users can view the current Fear and Greed Index, which is calculated based on factors such as market volatility, momentum, price strength, and investor behaviorproviding insight into overall market sentiment.

- **Key Events**
  - Users can view recent key events affecting the market, including detailed descriptions and their level of impact.

- **Technical Indicators**
  - Users can view current market technical indicators, including RSI, MACD, and Bollinger Bands.

8.2.3 Risk Signal Page

This module processes historical or real-time financial data and automatically generates exchange rate risk signals.

- **Parameter Settings**
  - **Currency Pair:** A dropdown menu allows users to select different currency pairs (e.g., USD/CNY, EUR/USD) to specify the target pair for analysis.

- **Risk Signal Strength**
  - Users can click to view the current risk signal strength.

- **Historical Signal Trend**
  - Users can view the trend of risk signals over a past period (e.g., past 30 or 90 days).

- **Forecast Analysis**
  - Users can view the 24-hour risk forecast, including risk scores, confidence levels, and predicted risk categories.

- **Factor-Based Risk Warning**
  - Users can click to view the fluctuation trends and impact levels of various contributing factors.

8.2.4 Backtesting Results Page

This module displays the backtesting performance of risk signals under various market conditions.

- **Backtesting Period**
  - **Maximum Drawdown:** A slider allows users to set the maximum drawdown percentage.
  - **Trading Period:** A numeric input box allows users to define the trading duration (e.g., 14 days).
  - **Dropdown Menu:** Users can select different backtesting durations (e.g., 1 year).
- **Yield Curve**
  - Users can click on the yield curve to explore performance over time.
- **Key Metrics**
  - Users can click each metric for detailed explanations, such as annualized return, max drawdown, Sharpe ratio, and win rate.
- **Trading Statistics**
  - Users can view detailed statistics such as total number of trades, profit/loss ratio, and average holding period.
- **Hedging Strategy**
  - Users can adjust the weights of currency pairs in the hedging strategy by clicking on them.

8.2.5 Multi-Currency Pair Analysis Page

This module analyzes multiple currency pairs based on their correlations and generates relevant risk signals.

- **Currency Pair Correlation Matrix**
  - Users can view correlation coefficients between different currency pairs in a matrix format.
- **Portfolio Risk Analysis**
  - Users can view risk indicators and analyze their influencing factors.
- **Currency Pair Weight Distribution**
  - Users can define the weight of each currency pair in their portfolio.
- **Integrated Risk Signals**
  - Users can view risk signals for each currency pair.

8.2.6 Settings Page

- **Signal Parameter Settings**
  - **Volatility Threshold:** A slider that allows users to set a percentage value.
  - **Signal Period:** A numeric selector that allows users to choose the signal duration.
  - **Confidence Level:** A dropdown menu that allows users to select a confidence level.
  - **Time Interval:** A dropdown menu that allows users to choose the time interval.
- **Data Source Settings**
  - **Data Source:** A dropdown menu for selecting the data source (e.g., real-time data, simulated data).
  - **Auto Refresh:** A toggle switch to enable or disable automatic refresh.
  - **Refresh Interval:** A slider to set the auto-refresh interval (e.g., every 5 minutes).
- **Currency Pair Selection**
  - By clicking 0 selected, a list will expand allowing users to choose one or more currency pairs.
- **Alert Settings**
  - **Enable Alerts:** A toggle to turn alert functionality on or off.
  - **Push Notifications:** A toggle to choose whether to receive push notifications.
  - **Email Notifications:** A toggle to choose whether to receive email notifications.
  - **Risk Threshold:** A slider to set the risk threshold (e.g., 70%).
  - **Notification Delay:** A slider to set the delay time for alerts.
  - **Alert Sound:** A dropdown menu to choose a notification sound.
  - **Custom Alert Rules:** Users can add new alert rules or edit existing ones.
- **Data Management**
  - **Clear Cache:** A button to clear the app cache.
  - **Export Settings:** A button to export the current settings.
  - **Reset All Settings:** A button to reset all settings to their default values.

8.2.7 Forex Data Search Page

- **Search Bar**
  - **Search Box:** Users can enter search content, such as USD/CNY exchange rate trend analysis or EUR exchange rate forecast.
  - **Search Button:** Clicking the button will trigger a search operation.
  - **Set API Key Button:** This feature allows users to manually set or update the API key used for external API calls. The platform has already completed backend integration with the DeepSeek API and configured an initial key, enabling users to use the service for free. For further customization or key updates, users may use this button.
- **Search Result Display Area**
  - **Result List:** Search results are presented in a list format. Each result item includes key information such as:
    - **Title:** Indicates the core content of the result.
    - **Reason:** Explains why the result matches the users query.
  - **Result Details:** After results are shown, users can click to bookmark or share the result.

9 Configuration Requirements

9.1 Python

Ensure that Python is fully installed on your machine and that the appropriate compiler environment is properly configured.

9.2 Python Packages

If you encounter errors such as the example below when running the code, it indicates that your system is missing required Python libraries. Please use `pip` to manually install the necessary packages to ensure the code runs correctly. (For the specific Python libraries required by the backend files, please refer to 2 and 3.)

Example Error Message:

```
ModuleNotFoundError: No module named 'sklearn'
```

Installation Command: Run the following command in the terminal or command prompt to install the required Python libraries:

```
pip install scikit-learn pandas sklearn shap
```

Descriptions:

- `scikit-learn`: A library for machine learning and data analysis.
- `pandas`: A library for data processing and analysis.
- `shap`: A library for interpreting the output of machine learning models.

10 Technical Support

- **Error Code Reference Table:**

Code	Description
1001	Database connection timeout
2004	API quota exceeded